

# 11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

July 9, 2026

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

**The business decision.** People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set the right **budget** — the most we can pay per exposure and still come out ahead.

### 1.0.1 The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer’s unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05’s adjustment tools are powerless here.

### 1.0.2 The fix: an instrument

An **instrumental variable (IV)** is a third variable  $Z$  that nudges the treatment *without* affecting the outcome any other way. Here  $Z$  is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because  $Z$  is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs three things:

- **Relevance** —  $Z$  actually moves exposure (this one we *can* test: the **first-stage F-statistic**;  $F < 10$  is a “weak instrument” danger sign).
- **Exclusion** —  $Z$  affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** —  $Z$  is independent of the hidden confounder (guaranteed here by randomization).

### 1.0.3 What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the “compliers,”* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

**On real data.** IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement** / **intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument.

## 1.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved **engagement** drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized **encouragement** (the instrument) nudges exposure without touching sales directly.

TRUE exposure->sales effect €15 · first-stage: exposure 56%->77% (shift +21%),  
F = 156

|   | encouragement | ad_exposure | sales     |
|---|---------------|-------------|-----------|
| 0 | 1.0           | 0.0         | 63.748992 |
| 1 | 1.0           | 1.0         | 64.636639 |
| 2 | 1.0           | 1.0         | 46.507627 |
| 3 | 0.0           | 1.0         | 75.599770 |
| 4 | 0.0           | 1.0         | 70.577754 |

## 1.2 3 · Identify — a valid instrument, and what IV recovers

Exposure is **endogenous** ( $\text{Cov}(\text{exposure}, \varepsilon) \neq 0$ ), so OLS is biased. A valid instrument  $Z$  (encouragement) needs three things:

- **Relevance** —  $Z$  moves exposure. *Testable* (the first-stage F above;  $F < 10$  = weak).
- **Exclusion** —  $Z$  affects sales *only through* exposure. *Untestable* — defended on design (a serving lottery sells nothing by itself). Stress-tested in step 6.
- **Exogeneity** —  $Z$  independent of the confounder. Holds by randomization.

**What IV identifies — LATE (Imbens–Angrist):** under monotonicity (no defiers), IV recovers the effect for **compliers** — customers the instrument moves — not everyone. CausalPy fits a Bayesian IV: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity  $\rho$  rather than assuming it away.

## 1.3 4 · Estimate — OLS, reduced form, and Bayesian IV

Three regressions tell the whole story, and seeing them side by side demystifies what IV actually does:

- **OLS** — regress sales on exposure directly. This is the **biased** number, inflated by self-selection.

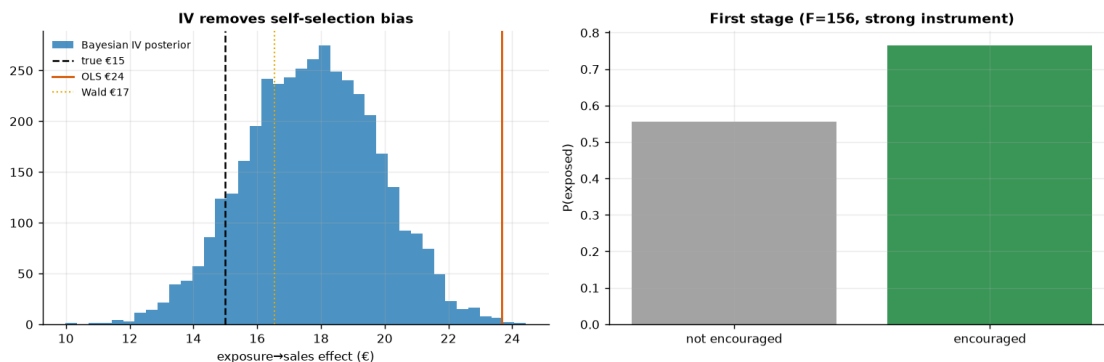
- **Reduced form** — regress sales on the *instrument* (encouragement -> sales). This captures only the effect that flows through the clean, randomized channel.
- **First stage** — regress exposure on the instrument (encouragement -> exposure): how much the nudge moves exposure.
- **Wald / IV ratio** — reduced form  $\div$  first stage. The intuition: the instrument raised sales by *this* much and raised exposure by *that* much, so each unit of (instrument-driven) exposure is worth their ratio. That ratio *is* the causal effect. The **Bayesian IV** below is the same idea fit jointly with uncertainty (and it estimates the endogeneity  $\rho$  rather than assuming it away).

OLS (biased)                      €23.7  
 Reduced form (Z->Y) €3.5  $\div$  first stage (Z->D) 0.21 = Wald/IV ratio €16.5  
 Bayesian IV                      €17.7 (true €15) 90% CI [€14.3, €21.1]  
 IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0

#### 1.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior’s 90% interval **covering the truth**. The gap between OLS and IV **is** the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy’s default 2SLS-centred priors would produce.)

self-selection bias removed: OLS €23.7 -> IV €17.7 (true €15)



**How to read this.** *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a little finite-sample noise (a euro or two above 15), but its posterior — now honestly wide, thanks to weakly-informative priors — comfortably **covers €15**; the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of “effect” that were never causal. If you budgeted on OLS you’d over-spend on every exposure by exactly that gap. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in P(exposed)), which is why the instrument is **strong** ( $F \gg 10$ ).

A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

## 1.5 6 · Decide, in euros — and the exclusion-restriction stress test

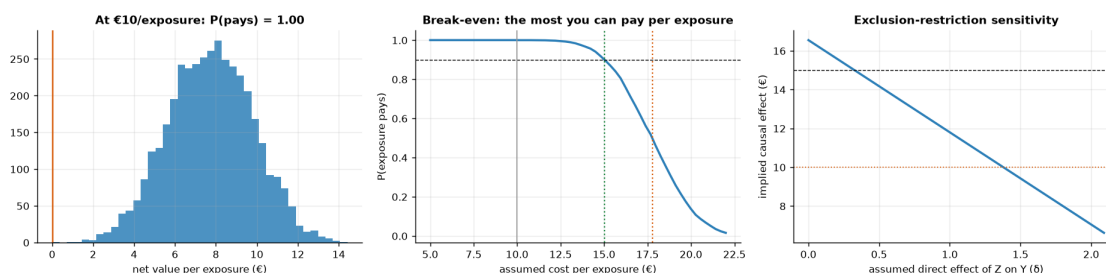
Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect  $\delta$  from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

At €10/exposure: causal net €7.7 · P(pays) 1.00 → BUY exposure (to the complier margin)

P(pays)=1.00 is saturated, not assumed: the whole posterior [€14.3, €21.1] sits above €10.

Break-even sweep: exposure clears the 90%-confidence bar up to €15.0/exposure (the budget cap); at €17.8 it is a coin flip.

So the €10 plan has ~€5.0 of headroom per exposure before the call gets tight. Exclusion stress: the BUY call survives unless the instrument's direct effect exceeds  $\delta \sim 1.4$  (out of a €3.5 reduced form) — trusting OLS (€24) would overstate the value and overspend.



## 1.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves. Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first-stage (low F) inflates variance and bias. We checked F above; a weak instrument is worse than none.
- **Exclusion is untestable.** We stress-tested it in step 6; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating.